

In India...

► "At the moment, the market is at an early stage due to cost economics, but we see that changing as innovation drives consumer value and competition drives price down for mass market adoption. The other trend that will help is the real estate industry increasingly building smart technology into homes and innovative service providers bringing 'solutions' to consumers by integrating these smart gadgets with telecom, mobile and tablets. Current devices, we are already seeing in reasonable numbers, include CCTV surveillance systems, sensor-based LEDs and climate control apps on smartphones."—**Guru Ganesan, president and MD, ARM India**

► "In India, the idea of smart home is slowly catching up with luxury homes and corporate offices making a conscious effort towards providing green environments. The idea is yet to become pervasive amongst the common man and may be incentives to support green initiatives and lower costs will increase the adoption. There is also growing interest in big cities as it does offer a lot of safety, security and automation benefits.

"For instance, it does enable working couples to manage their homes much more efficiently, help remotely monitor old-age parents and respond to any issues in a timely and effective manner. The most popular home automation systems today are safety and security systems, automated lighting, automated entertainment and automated energy management amongst others."—**Thilak Kumar, regional engineering manager, Wind River**

► "In India, smart homes have gone from being a premium concept to one that is fast becoming mainstream. Realtors have been integrating this into their constructions for a while now, and the

average home owner is increasingly looking for ways to automate. The last two years have seen cost of technology and the speed of its advancement become inversely proportionate.

"As technology advances, costs continue to come down and affordability is therefore driving smart home adoption. Its application has evolved as well. In that, home automation has moved from switching buttons on and off, to complete home surveillance and helping to operate and observe the home remotely. Intelligent fire safety systems, gas leakage detectors, motorised curtains, remote controlled lighting, video phone, automated alarms, remote monitoring of homes via Internet and adjustment of heating/cooling, wireless door locks and video door stations for security are some of the systems that have gained popularity. Again, as these become mainstream, we are of the firm opinion that smartphones will remain at the centre of driving adoption as they are being used as the primary controllers of these automation technologies."—**Annie Mathew, director, Alliances and Business Development, BlackBerry India**

► "Smart home as a concept is on the rise in India. With increased need for security, builders and real-estate developers are installing holistic surveillance systems that help to monitor large residential campuses. In India, cameras with smarter analytics are witnessing growing demand, especially the vandal-proof cameras and video motion detectors. Here, one can also witness higher adoption of thermal cameras, cameras with HD motion sensors and facial recognition software."—**Sudhendra Holla, country manager, Axis Communications, India**

smarthome solutions today, such as the Samsung Smart Home platform, Canary, Insteon Hub, MobiLinc, Roger Smart Home Monitoring and so on.

Roger Smart Home Monitoring, for example, helps subscribers to view and control their smart home monitoring system from wherever a mobile Internet connection can be made. The app helps to remotely arm and disarm your smart home monitoring system, and allows you to control security and monitoring devices including cameras, thermostats and lights.

Technologies such as NFC and Miracast make it easy to wirelessly connect a mobile phone to the television, home theatre or sound system, and enable users to stream their mobile games to the television or music from the mobile radio to the home speaker systems.

An app that comes to mind at this juncture is NFC Shortcuts, which allows users to launch apps or shortcuts conveniently by tapping a BlackBerry phone to NFC tags. "For example, when you reach home, you can just tap an NFC tag at the entrance and your BlackBerry will turn off the cellular radio, turn on Wi-Fi (home network),

turn off Bluetooth (used for hands-free in the car) and even send an SMS to your family members (not at home) notifying them that you are home. You can even tap a tag on your nightstand to set the BlackBerry into bedside mode and tap it again in the morning to take the device out of bedside mode," explains Mathew.

Mobile phones enable the concept of smart home to extend beyond the home too. The SkyPark Café in Hyderabad is an exciting example. The cafe is set up with BlackBerry devices, NFC points and a BlackBerry music gateway to play music to customers in a jukebox style.

In addition to that, PlayBooks are used to browse, work and play on and also double as the menu. There is also a wall dedicated with NFC stickers for customers to easily check-in with NFC-compatible devices. And there are also QR codes that customers scan to avail of freebies as well as an App of the Month offer.

Plus, nowadays there are smartphone apps such as Control4, which integrate all your home automation systems from media, security, lights

and thermostats using one easy-to-use interface.

For example, a single 'goodnight' button could ensure all your lights are switched off and doors are locked. A 'cooking' button could brighten the kitchen lights, switch on the chimney, pre-heat the oven, and play your favourite music on the kitchen speakers! Life becomes even easier.

"Home automation has created opportunities for consumer durables to cross-sell devices and products that are compatible with each other. Smartphones have driven this trend, given that it is today our single view of the world and consumer durable brands are fast adopting capabilities such as NFC and Miracast to ensure that consumers are able to control their homes from their smartphones," says Mathew.

Do it yourself

Obviously, you know your needs best, and that is the greatest motivation for DIY in the home automation space. A simple web search and you will be overwhelmed by a mind-boggling list of projects in this space. Low-cost microcontrollers and development boards